# Project Proposal

Subject: Speech Analysis and Linguistics

Priyanshu Panchasara (2026701004)
Vibhav Pandey (2026701006)

---

## 1 Title, Conference and Publication Year

**Title:** Self-Supervised Speech Representations are More Phonetic than Semantic

**Conference:** Interspeech

**Publication Year:** 2024

**Link:** arxiv.org/abs/2406.08619

## 2 Objectives

The main aim of this project is to check whether speech models pay more attention to how a word *sounds* than to what it *means*. To do this, we have the following objectives:

1. **Build the word pairs.** Create pairs of words that sound alike but have different meanings, and pairs of words that have the same meaning but sound different. We also create random pairs and same-word pairs to use as a comparison.

2. **Measure how close the words are inside the models.** Use speech models like wav2vec 2.0, HuBERT, etc. and check, layer by layer, how similar the model's representations of the two words in a pair are.

3. **Compare sound and meaning.** Find out whether words that sound alike are placed closer together than words that mean the same, and whether both are closer than random words.

4. **Test how our choices change the result.** Check whether the findings stay the same when we change how the word is cut from the audio, how its features are combined into one vector, and whether the audio comes from many speakers or a single speaker.

5. **Check whether a common test really measures meaning.** Compare a simple word-counting method with the speech models on two voice-command datasets (Fluent Speech Commands and Snips Smartlights), to see if high scores on these datasets truly show that the models understand meaning.

# 3 Datasets Required, Availability and Download Status

All the datasets needed for this project are free and publicly available.

1. **LibriSpeech (dev-clean and test-clean) with word alignments**
   English speech data, along with the start and end times of each word, used to cut out single words.
   Word alignments: https://zenodo.org/record/2619474
   Speech (Hugging Face): https://huggingface.co/datasets/openslr/librispeech_asr

2. **WordNet and CMU Pronouncing Dictionary**
   Word resources used to find words with the same meaning and words that sound alike.
   WordNet: downloaded through NLTK (`nltk.download('wordnet')`)
   CMU Pronouncing Dictionary: http://svn.code.sf.net/p/cmusphinx/code/trunk/cmudict/cmudict-0.7b

3. **Fluent Speech Commands and Snips Smartlights**
   Voice-command datasets (with their train/test splits) used to check whether high scores really show that the models understand meaning.
   Fluent Speech Commands (official): https://fluent.ai/fluent-speech-commands-a-dataset
   Snips Smartlights: https://huggingface.co/datasets/MWilinski/snips_slu_v1.0/tree/main

**Downloaded:** No

# 4 Linguistic Aspects and Analysis Plan

## 4.1 Linguistic Aspects

This project looks at how speech models represent words. We study the following linguistic aspects:

1. **Phonetics (how a word sounds).** Some words sound very similar even though their meanings are different, for example *dog* and *dig*. We find the sounds of each word using the CMU Pronouncing Dictionary and count how many sound changes are needed to turn one word into the other. Words with a small count are treated as sound-alike pairs.

2. **Semantics (what a word means).** Some words have a similar meaning but sound different, for example *dog* and *puppy*. We find such words using WordNet, a database that groups words by meaning.

3. **Word identity and speaker.** The same word spoken twice should look very similar to the model, so same-word pairs show the highest similarity we can expect. A speaker's voice can also affect the result, so we compare different words from the same speaker to see how much the voice matters.

4. **Context.** The words around a word can change how it is heard. We compare two ways of getting a word's representation: cutting the word out of the audio first, and using the whole sentence and then picking out the word.

5. **Meaning in use.** Voice assistants need to understand what a person wants, such as "turn on the kitchen lights". We check whether high scores on such tasks really show that a model understands meaning, or whether recognizing the words is enough.

## 4.2 Analysis Plan

We will carry out the analysis in the following steps:

1. Cut single words out of the English speech recordings using the word start and end times.

2. Make word pairs of five types: sound-alike, same-meaning, same-word, same-speaker and random.

3. Pass each word through the speech models like wav2vec 2.0, HuBERT, etc. and keep the output of every layer.

4. Measure how close the two words in each pair are, layer by layer, using cosine similarity.

5. Draw a graph for each model that shows the closeness of each pair type across the layers, and compare sound-alike pairs with same-meaning pairs.

6. On the voice-command datasets, train a simple decision tree that only uses which words appear in each command, and compare its accuracy with the speech models.

**Expected result:** We expect that words that sound alike will be closer to each other than words that mean the same, in the layers of all the models.

# 5 Project Execution Plan

## Phase 1: Understanding the Methodology

We study the paper in detail, especially the authors' methodology. We then prepare the working environment, and download and inspect the datasets.

## Phase 2: Building the Word Pairs

We cut single words out of the English recordings and build sound-alike pairs from their pronunciations and same-meaning pairs using WordNet. We also build same-word, same-speaker and random pairs for comparison.

## Phase 3: Implementing the Measurements

We pass the words through the speech models and measure how close the two words in each pair are, layer by layer, drawing graphs to compare sound-alike and same-meaning pairs. We then run the voice-command check, comparing a simple word-counting method with the speech models.

## Phase 4: Analysis, Report and Presentation

We compare our results with the findings of the paper and note any differences. We then write the final report and prepare the presentation.